

MEWC-LLM

Memory-Efficient Weight Consolidation for Large Language Models

Adaptive Stability–Plasticity Optimization for Factual Continual Learning in Large Language Models

A Research Proposal

Research Domain: Artificial Intelligence / NLP

Focus: Hallucination Reduction in LLMs via Continual Learning

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Abstract

Hallucination in large language models (LLMs) represents one of the most critical barriers to real-world deployment. Existing mitigation strategies — retrieval-augmented generation (RAG), replay buffers, and post-hoc fact-checking — treat hallucination as an inference-time artifact rather than a learning-dynamics failure. This paper introduces MEWC-LLM (Memory-Efficient Weight Consolidation for Large Language Models), a novel continual learning architecture that addresses hallucination at its root: the destabilization of factual knowledge representations during parameter updates.

MEWC-LLM integrates four interconnected mechanisms: (1) Truth-Aware Adaptive Consolidation using gradient cosine similarity to detect and suppress conflicts between new knowledge gradients and verified factual gradients; (2) Fisher-Based Truth Importance Scoring to protect parameters critical for factual reliability; (3) Conflict-Aware Truth Preservation (CATP) — an adaptive hallucination suppression layer that acts as a gradient-level immune system; and (4) a Modular Factual Memory Architecture separating stable reasoning from adaptable domain knowledge. The system operates without replay buffers, making it highly suitable for edge AI, private LLMs, and real-time continual learning environments.

Experimental validation on TruthfulQA, HaluEval, FEVER, MMLU, HotpotQA, and Natural Questions benchmarks is proposed. Expected outcomes include 40–70% standalone hallucination reduction in continual-learning scenarios and 80–95% reduction when combined with RAG and confidence calibration in bounded domains. MEWC-LLM represents a paradigm shift from external patching to internal truth preservation at the parameter level.

Keywords: LLM hallucination, continual learning, stability–plasticity dilemma, elastic weight consolidation, gradient conflict detection, factual memory, catastrophic forgetting, truth-preserving fine-tuning.

1. Introduction

The deployment of large language models (LLMs) across high-stakes domains — medicine, law, education, and scientific research — has exposed a fundamental reliability crisis: hallucination. LLMs routinely generate text that is fluent, contextually plausible, and factually incorrect. Unlike classical software bugs, hallucinations emerge from the statistical nature of neural language models and cannot be patched through simple rule-based fixes. They represent a deep architectural challenge rooted in how knowledge is encoded, retrieved, and updated within model parameters.

The scale of this problem is significant. Benchmark evaluations have consistently shown that state-of-the-art LLMs hallucinate at rates between 15% and 52% depending on task domain and model size. Even the most powerful frontier models — GPT-4, Claude, Gemini — regularly fabricate citations, misattribute facts, and generate plausible-sounding but unverifiable claims. In safety-critical applications, even a 5% hallucination rate is unacceptable.

Current mitigation approaches fall into two broad categories. External patches, such as Retrieval-Augmented Generation (RAG), ground model outputs in external documents at inference time. Internal patches, such as reinforcement learning from human feedback (RLHF) and chain-of-thought prompting, attempt to steer model behavior through training signals or reasoning scaffolds. While these approaches yield measurable improvements, they share a common limitation: they do not address the root cause of hallucination at the level of learning dynamics.

This paper proposes a fundamentally different framing. We argue that hallucination in LLMs is not primarily an inference-time retrieval failure but a learning-dynamics failure — specifically, a manifestation of the classical stability–plasticity dilemma in continual learning. When a model is fine-tuned on new data, gradient updates that encode new knowledge can overwrite or destabilize

internal representations of verified factual knowledge. This phenomenon, known as catastrophic forgetting in the continual learning literature, directly produces hallucination: the model loses reliable access to facts it once represented correctly.

Building on this insight, we introduce MEWC-LLM: Memory-Efficient Weight Consolidation for Large Language Models. MEWC-LLM extends and adapts the Elastic Weight Consolidation (EWC) framework — originally developed for task-based continual learning — to the challenge of factual truth preservation in LLMs. The system introduces gradient conflict detection, Fisher-based truth importance scoring, adaptive consolidation, and a modular factual memory architecture. Crucially, it achieves these properties without replay buffers, making it computationally tractable and suitable for deployment at scale, including on edge devices and in private, air-gapped LLM deployments.

The remainder of this paper is structured as follows: Section 2 defines the problem formally. Section 3 reviews related work. Section 4 states research questions. Section 5 presents objectives. Section 6 details the MEWC-LLM methodology. Section 7 describes the experimental setup and expected results. Section 8 discusses outcomes and impacts. Section 9 concludes. References follow.

2. Problem Statement

2.1 The Stability–Plasticity Dilemma in LLMs

Neural networks face an inherent tension between two competing requirements: stability (the preservation of previously learned knowledge) and plasticity (the capacity to acquire new knowledge). In biological neural systems, this tension is resolved through neuromodulatory mechanisms and complementary learning systems. Artificial neural networks, and LLMs in particular, lack these built-in regulatory mechanisms.

When an LLM undergoes continual fine-tuning — a common requirement in production deployments where models must adapt to evolving domain knowledge, updated factual information, or user-specific requirements — gradient updates for new tasks can catastrophically interfere with parameters encoding prior factual knowledge. The result is a model that is more plastic (adaptable) but less stable (reliable). This trade-off manifests directly as hallucination.

2.2 Formal Problem Definition

Let θ denote the parameters of a pre-trained LLM. Let $F = \{f_1, f_2, \dots, f_n\}$ represent the set of verified factual propositions reliably encoded in θ prior to fine-tuning. Let D_{new} represent a new training dataset. The fine-tuning process updates $\theta \rightarrow \theta'$ via gradient descent to minimize loss $L(D_{\text{new}}; \theta)$.

The hallucination problem in continual learning arises when the update $\theta \rightarrow \theta'$ degrades the model's ability to faithfully represent and retrieve facts from F . Formally, hallucination occurs when:

Hallucination condition: $P(\theta' \text{ correctly represents } f_i) < P(\theta \text{ correctly represents } f_i)$ for some $f_i \in F$

The central challenge is to identify which parameters in θ are critical for representing F , and to constrain gradient updates such that these parameters are protected during fine-tuning — without sacrificing the model's ability to learn from D_{new} .

2.3 Why Existing Approaches Are Insufficient

Current approaches address symptoms rather than causes:

Approach	Fundamental Limitation
Retrieval-Augmented Generation (RAG)	External patch; does not prevent internal representation degradation; fails when retrieved documents are incorrect or unavailable
RLHF / Preference Optimization	Shapes output distribution but does not protect specific factual parameters; expensive and requires continuous human feedback
Replay Buffers	Computationally expensive; privacy-violating; not feasible for edge or private deployments; does not scale to LLM parameter counts
Chain-of-Thought Prompting	Inference-time scaffold; does not address underlying parameter degradation; model can still hallucinate within reasoning chains
Post-hoc Fact-Checking	Reactive, not preventive; cannot verify all claims; requires reliable external ground truth

3. Related Work

3.1 Hallucination in Large Language Models

The literature on LLM hallucination has grown substantially since 2022. Ji et al. (2023) provided an early comprehensive survey defining hallucination types: intrinsic (contradicting the source) and extrinsic (unverifiable against the source). Maynez et al. (2020) demonstrated that faithful summarization is fundamentally distinct from fluent summarization, establishing that neural models can be highly fluent while being factually unreliable.

Mallen et al. (2023) showed that LLMs systematically hallucinate more on less-popular factual queries, suggesting that hallucination is not random but structured — tied to the frequency and consistency of factual representations in training data. Xu et al. (2024) provided a formal proof that hallucination cannot be fully eliminated under current LLM architectures, establishing the theoretical ceiling that motivates bounded-reduction approaches such as MEWC-LLM.

3.2 Continual Learning and Catastrophic Forgetting

The catastrophic forgetting problem in neural networks was first described by McCloskey and Cohen (1989) and later formalized by Ratcliff (1990). Kirkpatrick et al. (2017) introduced Elastic Weight Consolidation (EWC), which uses the Fisher information matrix to identify and selectively protect important parameters during sequential task learning. EWC has since become a foundational method in the continual learning literature.

Subsequent work has extended EWC in multiple directions. Online EWC (Schwarz et al., 2018) reduces memory requirements by maintaining a running estimate of Fisher importance. Progress & Compress (Schwarz et al., 2018) combines EWC with knowledge distillation. PackNet (Mallya and Lazebnik, 2018) uses parameter masking to achieve zero forgetting at the cost of architectural rigidity. None of these methods have been adapted specifically to the challenge of factual truth preservation in LLMs.

3.3 Gradient Conflict Detection

The problem of conflicting gradients in multi-task learning was analyzed by Yu et al. (2020), who showed that gradient conflicts between tasks are a primary driver of negative transfer and forgetting. PCGrad (Yu et al., 2020) projects conflicting gradients onto orthogonal subspaces to reduce interference. GradNorm (Chen et al., 2018) adaptively weights task gradients to balance

learning dynamics. CAGrad (Liu et al., 2021) finds gradient updates that minimize the worst-case loss across tasks.

These methods, while effective in multi-task settings, have not been applied to the problem of factual knowledge preservation in LLMs. MEWC-LLM adapts the core insight of gradient conflict detection — that conflicting gradient directions indicate knowledge interference — to the specific context of truth preservation.

3.4 Retrieval-Augmented Generation

Lewis et al. (2020) introduced RAG as a method for grounding LLM outputs in retrieved documents. Subsequent work has extended RAG with dense retrieval (Karpukhin et al., 2020), multi-hop reasoning (Yang et al., 2018), and adversarial verification (our HALO framework). While RAG is effective in controlled domains, it introduces latency, requires maintained retrieval infrastructure, and remains susceptible to retrieval failures and document-level hallucinations.

3.5 Fisher Information in Neural Networks

The Fisher information matrix (FIM) quantifies the sensitivity of a model's output distribution to changes in its parameters. Martens (2014) showed that the FIM provides a natural metric for parameter importance in the context of natural gradient descent. Kirkpatrick et al. (2017) exploited this property for continual learning. Computing the full FIM is computationally intractable for LLMs; diagonal approximations and low-rank decompositions have been proposed (Ritter et al., 2018; Immer et al., 2021).

3.6 Modular Architectures for Knowledge Isolation

Mixture-of-Experts (MoE) architectures (Jacobs et al., 1991; Shazeer et al., 2017) provide a structural basis for separating knowledge across specialized modules. More recently, LoRA (Hu et al., 2022) demonstrated that low-rank adaptations can achieve effective fine-tuning with minimal parameter overhead. Adapter-based methods (Houlsby et al., 2019) offer a related approach. MEWC-LLM's Modular Factual Memory Architecture builds on these foundations with an explicit focus on truth-preserving isolation.

4. Research Questions

This research is guided by five primary research questions:

1. Can gradient cosine similarity between new-knowledge gradients and verified-factual gradients serve as a reliable signal for detecting hallucination-inducing parameter updates in LLMs undergoing continual fine-tuning?
2. Does adapting Fisher information-based weight consolidation — shifting the target from task preservation to factual truth preservation — yield measurable reductions in hallucination rates on established benchmarks without substantially degrading continual learning performance?
3. To what extent does a Conflict-Aware Truth Preservation (CATP) layer — operating as a gradient-level immune system with selective update rejection, learning rate modulation, and retrieval activation — reduce catastrophic forgetting of verified factual representations?
4. Does a Modular Factual Memory Architecture (shared reasoning backbone + domain-specific factual heads) reduce hallucination propagation compared to monolithic architectures in multi-domain continual learning scenarios?
5. What is the computational overhead of MEWC-LLM relative to standard fine-tuning, and is it feasible for deployment in resource-constrained environments (edge AI, private LLMs) where replay buffers and large retrieval systems are unavailable?

5. Objectives

5.1 Primary Objectives

1. Develop and formalize the MEWC-LLM framework: a no-replay, gradient-conflict-aware continual learning architecture for hallucination-resistant LLMs.
2. Formalize the Truth-Aware Adaptive Consolidation (TAAC) mechanism using cosine similarity between factual and new-knowledge gradients.
3. Adapt Fisher Information Matrix computation to factual truth importance scoring, enabling selective parameter protection for LLM-scale models.
4. Design and implement the Conflict-Aware Truth Preservation (CATP) layer with gradient-level immune system semantics.
5. Architect the Modular Factual Memory system with domain-isolating factual heads and verification adapters.

5.2 Secondary Objectives

- Validate MEWC-LLM on six established hallucination and factual consistency benchmarks.
- Compare MEWC-LLM against EWC, RAG, RLHF, and no-mitigation baselines.

- Characterize the computational overhead of each MEWC-LLM component to establish feasibility profiles for edge and private deployment.
- Investigate the composability of MEWC-LLM with external approaches (RAG, confidence calibration, tool use) toward a comprehensive anti-hallucination system.
- Contribute open-source implementations, benchmark datasets, and evaluation tooling to the research community.

6. Methodology

6.1 Theoretical Foundation: The MEWC-LLM Objective

MEWC-LLM frames factual continual learning as a constrained optimization problem. The training objective is a modified loss function that balances adaptation to new data against protection of factual parameters:

$$\text{MEWC-LLM Loss: } L_{\text{MEWC}}(\theta) = L_{\text{new}}(\theta; D_{\text{new}}) + \lambda \cdot \sum_i F_i^{\text{truth}} \cdot (\theta_i - \theta_i^*)^2$$

where L_{new} is the standard training loss on new data D_{new} , F_i^{truth} is the Truth Importance Score for parameter θ_i (a Fisher-based estimate of how critical parameter i is for factual reliability), θ_i^* represents the parameter values encoding verified factual knowledge, and λ is an adaptive consolidation coefficient controlled by gradient conflict detection.

6.2 Component 1: Truth-Aware Adaptive Consolidation (TAAC)

The TAAC mechanism detects when new-knowledge gradient updates would conflict with factual knowledge gradients and adaptively adjusts the consolidation strength in response. Let g_t be the gradient vector for the current training batch, and g_{prev} be the gradient vector computed on a small set of verified factual examples (the Factual Anchor Set, FAS). The conflict signal is measured by cosine similarity:

$$\text{Conflict Signal: } \text{conflict} = -\cos(g_t, g_{\text{prev}}) = -(g_t \cdot g_{\text{prev}}) / (||g_t|| \cdot ||g_{\text{prev}}||)$$

A negative cosine similarity indicates that the new-knowledge update would move parameters in a direction that conflicts with factual knowledge preservation. The adaptive consolidation coefficient λ is then modulated:

$$\text{Adaptive Lambda: } \lambda_{\text{adaptive}} = \lambda_{\text{base}} \cdot (1 + \alpha \cdot \max(0, \text{conflict}))$$

where α is a sensitivity hyperparameter controlling how aggressively consolidation increases upon conflict detection. When conflict is high, $\lambda_{\text{adaptive}}$ increases, imposing stronger constraints on parameter updates near factual representations.

6.3 Component 2: Fisher-Based Truth Importance Scoring

The Fisher Information Matrix provides a principled measure of parameter importance. For parameter θ_i , the Fisher importance F_i quantifies how sensitively the model's output distribution changes with respect to small perturbations in θ_i . Standard EWC computes F_i with respect to a previous task. MEWC-LLM shifts this computation to a Factual Anchor Set (FAS) — a curated collection of verified factual query-answer pairs:

Truth Importance: $F_i^{\text{truth}} = E_{(x,y) \in \text{FAS}} [(\partial \log p(y|x; \theta) / \partial \theta_i)^2]$

Computing the full FIM is $O(n^2)$ in parameter count and intractable for billion-parameter LLMs. We adopt a diagonal approximation augmented with a layer-wise relevance propagation scheme to identify which layers carry the highest factual importance, enabling efficient computation. Specifically, we compute full diagonal Fisher for the top-K highest-importance layers and zero-approximate for the remainder.

6.4 Component 3: Conflict-Aware Truth Preservation (CATP) Layer

The CATP layer operates as a gradient-level immune system, evaluating gradient directions before parameter updates are applied. It implements a three-tier response protocol based on conflict severity:

Conflict Level	Threshold	CATP Response
Low conflict	$\cos(g_t, g_{\text{prev}}) > -0.1$	Normal update proceeds; standard λ applies
Moderate conflict	$-0.5 < \cos \leq -0.1$	Learning rate scaled down; $\lambda_{\text{adaptive}}$ activated; soft gradient projection

High conflict	$\cos \leq -0.5$	Update for factual parameters rejected; retrieval augmentation triggered; conflict logged
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The high-conflict rejection mechanism is analogous to an immune system response: detecting a potentially destabilizing update and blocking it before it damages established factual representations. When retrieval is triggered, the model is directed to ground its response in retrieved evidence rather than relying on potentially destabilized internal representations.

6.5 Component 4: Modular Factual Memory Architecture

MEWC-LLM introduces a shared-bottom architecture that structurally separates stable reasoning from adaptable domain knowledge. The architecture consists of three layers:

- **Shared Reasoning Backbone:** A frozen or highly constrained lower transformer stack encoding syntactic reasoning, logical inference, and cross-domain semantic representations. Protected by maximum TAAC consolidation.
- **Domain-Specific Factual Heads:** Lightweight adapter modules attached at the upper transformer layers, each specialized for a specific domain (medicine, law, science, etc.). Fine-tuning targets these adapters, isolating domain adaptation from core reasoning.
- **Verification Modules:** Dedicated attention heads trained to assess factual consistency between generated content and retrieved evidence. These modules output confidence scores that gate the CATP layer's response thresholds.

This modular design limits hallucination propagation: errors introduced in one domain adapter cannot corrupt the shared reasoning backbone or other domain adapters, significantly reducing the blast radius of any individual fine-tuning event.

6.6 No-Replay Truth Retention

Unlike replay-based continual learning methods, MEWC-LLM does not store or replay past training examples. Factual memory is preserved entirely through the F_i^{truth} importance weights and the Factual Anchor Set (FAS). The FAS is a compact, curated set of verified factual examples — typically 1,000–5,000 query-answer pairs per domain — used only for Fisher computation and gradient conflict detection, not for replay training.

MEWC-LLM vs Replay-Based Methods

Replay requirement: MEWC-LLM requires 0 replay examples; standard replay requires 10–50% of original training data

Memory overhead: MEWC-LLM adds $\sim 2\times$ parameter storage (for Fisher weights) + small FAS; replay requires full past dataset storage

Privacy compliance: FAS contains verified public facts, not user or training data; suitable for GDPR/HIPAA-constrained deployments

Edge feasibility: FAS and Fisher weights fit on consumer hardware; replay buffers for LLMs require datacenter-scale storage

7. Experimental Setup & Expected Results

7.1 Experimental Setup

Datasets and Benchmarks

Benchmark	Purpose in MEWC-LLM Evaluation
TruthfulQA	Primary hallucination rate measurement; 817 adversarially crafted questions targeting common LLM misconceptions
HaluEval	Hallucination detection at the claim level; tests whether MEWC-LLM correctly identifies and suppresses fabricated content
HotpotQA	Multi-hop factual reasoning; tests whether TAAC preserves complex reasoning chains across updates
MMLU	57-domain knowledge breadth; measures whether MEWC-LLM's factual protection generalizes across domains
FEVER	Fact verification; evaluates whether MEWC-LLM's internal representations align with verifiable fact databases
Natural Questions	Open-domain QA; measures practical factual retrieval fidelity after continual fine-tuning

Baselines

- No mitigation: Standard fine-tuning with no continual learning or hallucination controls
- EWC: Standard Elastic Weight Consolidation (Kirkpatrick et al., 2017)
- RAG-only: Dense retrieval-augmented generation without internal parameter protection
- LoRA fine-tuning: Parameter-efficient adaptation without factual protection
- MEWC-LLM (ablations): Individual components enabled/disabled to isolate contributions
- MEWC-LLM + RAG + Confidence: Full combined system (upper-bound estimate)

Models

Primary experiments will be conducted on LLaMA-3-8B and Mistral-7B-v0.3 (open-source, reproducible). Extended experiments on GPT-4-class models via API will validate scaling behavior.

7.2 Key Metrics

Metric	Definition
Hallucination Rate (HR)	Fraction of generated factual claims that are incorrect or unverifiable, measured on benchmark gold labels
Factual Consistency Score (FCS)	BERT-score-based alignment between generated content and verified source documents
Catastrophic Forgetting Index (CFI)	Performance degradation on pre-fine-tuning benchmark tasks after MEWC-LLM continual update cycles
Continual Adaptation Quality (CAQ)	Performance on new-domain tasks after fine-tuning; measures that factual protection does not block new learning
Conflict Detection Precision	Fraction of CATP-flagged gradient conflicts that correspond to actual factual degradation events
Computational Overhead Ratio	MEWC-LLM wall-clock training time relative to standard fine-tuning baseline

7.3 Expected Results

Based on theoretical analysis and preliminary analogies from EWC and gradient conflict detection literature, we project the following outcomes:

Configuration	Expected HR Reduction	Expected CAQ Change
TAAC alone (vs. standard fine-tuning)	20–35%	–3% to +2%
Fisher Truth Importance alone	15–25%	–4% to +1%
CATP layer alone	25–40%	–2% to +3%
Full MEWC-LLM (all 4 components)	40–70%	–5% to +4%
MEWC-LLM + RAG + Confidence	80–95% (bounded domains)	Neutral to +6%
MEWC-LLM vs. EWC (same compute)	MEWC-LLM –15–25pp lower HR	Comparable

The modest negative impact on CAQ (–5% in worst case) reflects the inherent stability–plasticity trade-off: stronger factual protection constrains some degree of new learning. MEWC-LLM's CATP layer is designed to minimize this trade-off by selectively blocking only high-conflict updates, preserving plasticity for non-conflicting new knowledge.

8. Outcomes & Impacts

8.1 Scientific Contributions

6. A formal theoretical framework connecting LLM hallucination to the stability–plasticity dilemma in continual learning — the first such formalization in the literature.
7. The MEWC-LLM architecture: a complete, reproducible, open-source system for factual truth preservation in continual LLM fine-tuning.
8. Truth Importance Scoring: an adaptation of Fisher information computation specifically targeting factual reliability in LLMs, distinct from task-based importance measures.
9. Conflict-Aware Truth Preservation (CATP): a novel gradient-level immune system mechanism with three-tier adaptive response, applicable beyond LLMs to any continual learning setting where specific knowledge must be protected.

10. Empirical benchmarks and evaluation protocols for measuring hallucination in continual learning scenarios, filling a gap in current evaluation methodology.

8.2 Practical Impacts

- Medical AI: MEWC-LLM enables clinical LLMs to adapt to updated treatment guidelines without losing reliable representations of established medical facts — directly reducing diagnostic hallucination risk.
- Legal AI: Contract analysis and legal research tools can incorporate new case law while preserving reliable representations of established legal precedent.
- Edge AI and Private Deployments: The no-replay, bounded-memory design makes MEWC-LLM the first factual continual learning framework explicitly designed for resource-constrained and privacy-sensitive LLM deployments.
- Robotics and Embodied AI: Physical systems requiring real-time knowledge updates (e.g., navigation, task completion) benefit from a model that adapts without forgetting safety-critical operational facts.

8.3 Toward Self-Correcting Lifelong LLMs

The ultimate vision of this research line is a self-correcting lifelong LLM: a system that continuously ingests new information, updates its internal representations, verifies its own factual claims against both internal and external sources, and refuses to generate content it cannot reliably ground. MEWC-LLM is a foundational component of this vision, providing the internal parameter-level protection layer upon which agentic verification, symbolic reasoning, and retrieval augmentation can be composed.

Combined with the HALO orchestration framework (Hybrid Anti-Hallucination LLM Orchestration), MEWC-LLM represents the internal learning-dynamics layer of a comprehensive anti-hallucination system stack. Where HALO provides inference-time verification and retrieval grounding, MEWC-LLM ensures that the base model's internal factual representations remain reliable across the model's operational lifetime.

9. Conclusion

This paper has presented MEWC-LLM: Memory-Efficient Weight Consolidation for Large Language Models, a novel continual learning architecture that addresses LLM hallucination at its root cause — the destabilization of factual knowledge representations during parameter updates. We have argued that hallucination is fundamentally a stability–plasticity failure, and that

effective long-term solutions must operate at the level of learning dynamics rather than relying solely on inference-time patches.

MEWC-LLM integrates four novel mechanisms: Truth-Aware Adaptive Consolidation using gradient cosine similarity for conflict detection; Fisher-Based Truth Importance Scoring for selective parameter protection; the Conflict-Aware Truth Preservation layer as a gradient-level immune system; and a Modular Factual Memory Architecture that structurally isolates stable reasoning from adaptable domain knowledge. Together, these components provide a principled, computationally tractable approach to factual truth preservation that requires no replay buffers and is suitable for edge and private LLM deployments.

We project 40–70% hallucination reduction in standalone continual learning scenarios and 80–95% reduction when combined with retrieval augmentation and confidence calibration in bounded domains. These results, if validated experimentally, would represent the most significant advance in hallucination reduction achieved through internal model mechanisms rather than external patching.

The framing of hallucination as a continual learning problem opens a rich vein of future research: adaptive factual anchoring, hierarchical truth importance, compositional factual memory, and the integration of symbolic verification with gradient-level protection. MEWC-LLM is a first step toward the ultimate goal of a trustworthy, self-correcting lifelong LLM — not a mathematically impossible hallucination-free system, but a reliably grounded one that refuses uncertainty, protects verified knowledge, and earns justified trust in high-stakes deployment.

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